Business Requirement Document (BRD)

1. Objective

Build iThink: a multi-tenant SaaS that ingests social media conversation data and produces attribute-based competitor benchmarking. First build target: the two-wheeler category, benchmarking TVS RTX 300 against Hero Xpulse 200, Royal Enfield Himalayan 450, and KTM 390 Adventure.

2. Background

Initially scoped against Hawky.ai (an ad-creative intelligence platform), the product direction shifted to a broader social/brand intelligence category once it became clear the desired capabilities — attribute benchmarking, consumer complaint analysis, executive monitoring — span beyond ad-performance tooling. Attribute-based competitor benchmarking was selected as the first buildable slice.

3. Scope

In scope (this build):

- Ingestion of social conversation data (Excel export today; Sprinklr/Brandwatch API and data-lake adapters later, same schema)
- Multi-label LLM tagging across three taxonomies: Business (8 domains), Consumer Attribute (168 attributes), AI Signal (80 tags: sentiment, emotion, intent, severity, importance, trend, impact, journey stage, competitive context)
- Brand identification per conversation (own brand vs. named competitors), including disambiguation of similarly-named but unrelated products (e.g., KTM Duke 390 vs. KTM 390 Adventure)
- Aggregation into brand × attribute × time-period benchmarking scores
- A dashboard presenting benchmarking views, filterable by attribute, brand, time period, sentiment
- Multi-tenant architecture with Row-Level Security and role-based access

Out of scope (backlog / future phases):

- Billing/subscription integration
- Defect Intelligence Engine (root-cause analysis on QAD-tagged conversations)
- Whitespace Engine (NPD-tagged demand aggregation)

- Executive/leadership monitoring, video/image complaint analysis, organic creative-content analysis (other capabilities considered earlier but not being built now)
- Change management, stakeholder communications, formal QA sign-off processes (excluded from this plan by request; this plan covers development & testing only)

4. Stakeholders

Role	Who	Interest
Product owner / developer	Arun	Builds and owns the product end-to-end
Future end users	Brand managers, market research teams at two-wheeler OEMs	Consume benchmarking dashboards

5. Functional Requirements

ID	Requirement
FR1	Ingest conversation data from Excel exports matching the Sprinklr export schema
FR2	Tag each conversation with one or more Consumer Attribute Taxonomy entries
FR3	Tag each conversation with one or more AI Signal Taxonomy entries
FR4	Identify which brand(s) a conversation refers to, disambiguating similarly-named unrelated products
FR5	Aggregate tagged conversations into brand × attribute × time-period scores
FR6	Present a dashboard with benchmarking views, filterable by attribute, brand, period, sentiment
FR7	Isolate tenant data via Postgres Row-Level Security
FR8	Support role-based access per tenant (admin / analyst / viewer)

6. Non-Functional Requirements

ID	Requirement
NFR1	All POC infrastructure runs on free-tier services
NFR2	Tenant data isolation enforced at the database level, not just application code
NFR3	Ingestion and extraction pipelines are idempotent (safe to re-run without duplicating data)
NFR4	Architecture accommodates new ingestion sources (Sprinklr/Brandwatch APIs) without schema changes

7. Assumptions

- Geography is inconsistently present in Sprinklr exports; stored as nullable, populated when the export provides it
- Language and Model/Variant are inferred by the LLM extraction pass; Sprinklr's own `Model/Variant` column is trusted when populated, LLM inference is the fallback
- Severity and Importance use the same Low/Medium/High scale as the existing Impact signal

8. Success Criteria

- End-to-end pipeline runs against real Sprinklr export data for RTX 300 and the three named competitors
- Dashboard renders at least one working benchmarking view (attribute × brand matrix)
- A second tenant can be onboarded without schema or code changes